

AI Box to Enterprise Gen AI Assistant

One Portable Codebase, Two Deployment Targets

GCP-Managed (default) vs No-Lock-In On-Prem

Decision-grade comparison for enterprise architects and the exec sponsor.

All figures INDICATIVE (2026): order-of-magnitude planning numbers, not vendor quotes.

Agenda

Part 1 — Overview. What we're building, the architecture, two topologies, framing assumption, locked decisions.

Part 2 — Deep Dive. Dimension by dimension: time-to-market, quality, cost/TCO, risk, security, sovereignty, scalability/ops.

Part 3 — The Decision. Weighted scorecard, sovereignty gate, risk register, where on-prem wins, rebuttal, portability caveats, roadmap, recommendation.

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Bottom Line Up Front

- **Build ONCE, deploy two ways.** One codebase; only adapters and the platform beneath them change.
- **Default to GCP-managed:** fastest TTM (~5× less assembly), best quality, lowest TCO (~35–40% of on-prem). Scorecard 4.67 vs 2.57 out of 5.
- **Build on-prem as insurance:** the no-lock-in / data-residency option; decisive only under a sovereignty mandate or already-sunk platform + people.
- **Not a one-way door:** portability is an architectural property of the hexagonal CORE, not a deployment choice.

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Part 1 of 3

Overview (high level)

Executive Summary

- **Build ONCE, deploy two ways.** Hexagonal (ports-and-adapters) design keeps a provider-agnostic CORE identical across both targets. Only the adapters and platform change.
- **Default to GCP-managed.** Best quality (Agent Search on Gemini Enterprise Agent Platform + Document AI + Gemini Flash), fastest TTM (~7 vs ~36 eng-weeks, ~5×), lowest TCO (~\$2.1M vs ~\$5.7M over 3 yrs).
- **TCO gap is people, not gear.** Inference is external in both. On-prem: ~4–6 FTE platform ops; GCP: ~1–1.5 FTE to integrate, not operate.
- **Honest baseline.** Year-0 ports-and-adapters refactor is costed once in both columns — that's why Year-0 is non-zero on both sides.
- **Build on-prem to a tested baseline as insurance.** No-lock-in / data-residency / air-gap option; competitive only when sovereignty mandates it or capacity is already sunk.

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What We Are Building

An **enterprise Gen AI document assistant**: upload documents, find what you're allowed to see, and chat over your files with cited answers.

Capability	What it does
Ingest	Upload PDF/text; parse, structure-aware chunk, index on arrival
Access-aware search	Hybrid keyword + semantic, ABAC-filtered; restricted matches show a redacted card
RAG chat with citations	Grounded only in docs the user may see; inline [n] citations; Validator groundedness gate
Access-request workflow	One-click request, approver email, time-limited grant
Audit	Append-only log of every auth, search, view, and AI turn

Security invariant: access enforced server-side on EVERY retrieval — never in the UI.

Architecture at a Glance: One Core, Swappable Adapters

Ports-and-adapters (hexagonal). The product is written once; the cloud underneath is a config choice.

Layer	Content	Changes per target?
CORE	ABAC, chunking, retrieval, ingest, audit; agent graph; /api routes; UI	No (byte-identical)
PORTS	LLM, Embedder, Reranker, Retriever, Store, ObjectStore, Parser, Guardrail, DLP, EventBus, Telemetry, Identity, Orchestrator	No (interfaces)
ADAPTERS	One implementation set per target	Yes
PROFILE	<code>AIBOX_PROFILE</code> = local / onprem / gcp selects the adapter set	The one switch

- ~80% of the code never changes between targets (CORE + UI + eval + tests).
- What differs: only `adapters/<target>/` , `profiles/<target>.yaml` , `deploy/<target>/` .

Deployment at a Glance: Two Topologies

Same request path; different platform beneath the ports. Inference is an external hosted endpoint in **both**.

GCP-managed: Apigee + Cloud Armor → Cloud Run (CORE) → Agent Runtime/ADK → Agent Search on Gemini Enterprise Agent Platform + Document AI over AlloyDB + GCS; Model Armor, Cloud DLP, Eventarc, Cloud Observability.

No-lock-in on K8s: Kong/Envoy + OIDC → CORE pod → ADK on K8s → OpenSearch + bge + Tika over pgvector + MinIO; Llama Guard + NeMo + Presidio, Knative/KEDA, OTel + Grafana.

Tier	GCP	On-prem
Gateway	Apigee X + Cloud Armor	Kong / Envoy + OIDC
Agent runtime	Agent Runtime on Gemini Enterprise Agent Platform + ADK	ADK on K8s
Retrieval	Agent Search on Gemini Enterprise Agent Platform + Document AI	OpenSearch + bge + Tika
Data	AlloyDB + GCS	pgvector + MinIO
Safety / DLP	Model Armor + Cloud DLP	Llama Guard + NeMo + Presidio

Framing Assumption: Inference Is External in BOTH Targets

No AI inference runs on-premise in either target. Both call a hosted model endpoint (GCP: Gemini on Gemini Enterprise Agent Platform; on-prem: Gemma via LiteLLM). The comparison is the PLATFORM around the model.

Implication	Consequence
Model inference \$ is equal both sides	Excluded from the differential TCO
No inference GPU on either side	On-prem gets no credit for "avoiding GPU spend"
Single model port is the seam	Swapping Gemini for Gemma is one adapter line, not a CORE change

The comparison is purely the platform: gateway, guardrails, agent runtime, retrieval, parsing, DB, storage, eventing, DLP, observability, and the people who run it.

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Scope and Assumptions

Area	Assumption
Workload	~500–2,000 internal users; ~1M docs; moderate bursty volume; ~99.9% target
Cost basis	~\$160k–\$280k/FTE loaded; managed services as usage-based opex; INDICATIVE (2026)
Inference	External hosted endpoint in both; cost equal, excluded from differential
In scope	Platform: gateway, agent runtime, retrieval, parsing, DB, storage, eventing, DLP, safety, observability + people
Out of scope	Model fine-tuning, GPU capex/ops, data-migration specifics, org compliance audits
Vendor figures	Context window, durability, infoType counts are vendor-cited as of 2026: verify at procurement

Locked Architecture Decisions

#	Decision (final)
1	Agent layer = ADK in both. GCP: Agent Runtime on Gemini Enterprise Agent Platform + ADK. On-prem: same ADK app on K8s. Agent graph + prompts live in CORE and are reused; only model binding and host differ.
2	Retrieval. GCP: Agent Search on Gemini Enterprise Agent Platform + Document AI; AlloyDB holds chunk text + ABAC side-tables. On-prem: OpenSearch (BM25+kNN) + bge-reranker; pgvector holds chunk text + ABAC.
3	Validator / groundedness gate runs before every answer, in both targets.
4	ABAC is CORE-owned (<code>AccessPredicate</code>), compiled per backend: SQL (SQLite/pgvector/AlloyDB), filter-DSL (Agent Search), DLS (OpenSearch). Policy model + orchestration reused; enforcement per-adapter.
5	Embedder and Reranker are first-class ports. On-prem depends on 4 external hosted inference endpoints (LLM, embedder, reranker, safety). GCP folds embedding + reranking into managed Agent Search on Gemini Enterprise Agent Platform.

Scaffold Reality (Honest Current State)

Profile	State today	What it proves
Prototype	SQLite + local FS + Anthropic API. No Postgres, no pgvector, no ADK graph, no port boundary yet.	Domain logic (ABAC, chunking, ACL retrieval, citations, audit) works end-to-end.
local	Runs end-to-end (lightweight in-process runner).	CORE prompts and Orchestrator graph are real and runnable.
onprem	Code-complete (docker-compose).	Four hosted-inference ports and OSS adapters are wired, not aspirational.
gcp	Real adapter code; requires a GCP project to run.	Gemini Enterprise Agent Platform / AlloyDB / Document AI adapters exist in code.

- The ports-and-adapters refactor is the shared Year-0 build, common to both targets. It nets out of the differential and explains the non-zero Year-0 on both sides.
- Reuse/portability claims are designed intent realized in this repo, not a claim that the GCP profile is in production.

Side-by-Side Stack Mapping

Everything above the ports ships unchanged (ABAC, chunking, citations, Validator, request/approval, audit, UI). Only the rows below differ.

CORE port	GCP-managed adapter	On-prem adapter
API gateway / edge	Apigee X + Cloud Armor	Kong / APISIX / Envoy + ModSecurity/Coraza
Safety / guardrails	Model Armor (injection + RAI + DLP-aware)	Llama Guard 4 + NeMo Guardrails + Presidio
Models / generation	Gemini on Gemini Enterprise Agent Platform	Gemma via hosted OpenAI-compatible endpoint (LiteLLM)
Agent runtime	Agent Runtime on Gemini Enterprise Agent Platform + ADK (managed)	Same ADK app on K8s (self-managed)
Retrieval + index + rerank	Agent Search on Gemini Enterprise Agent Platform (hybrid RRF + reranker)	OpenSearch (BM25+kNN) + bge-reranker-v2-m3
Doc parsing / OCR	Document AI Layout Parser	Tika / Unstructured + Tesseract/PaddleOCR
PII / DLP	Cloud DLP / Sensitive Data Protection	Presidio
Event-trigger ingestion	Cloud Functions + Eventarc	Knative / KEDA / Argo Events
Object store	GCS	MinIO (S3-compatible, erasure-coded)
Structured + vector store	AlloyDB (ScaNN, pgvector)	PostgreSQL 16 + pgvector (self-run HA)
Observability + eval	Cloud Observability + Gemini Enterprise Agent Platform Evals	OTel + Prometheus + Grafana + Langfuse/RAGAS/Phoenix

Stack Mapping: The Reuse Win

- **Embedder + Reranker are ports, not buried config.** On-prem must host them as separate inference endpoints; GCP folds both into managed Agent Search on Gemini Enterprise Agent Platform.
- **On-prem depends on 4 external hosted inference endpoints** (LLM, embedder, reranker, safety). GCP calls one model endpoint and bundles the rest.
- **One Postgres-compatible repository** serves both targets (AlloyDB is Postgres-wire-compatible). Same model port, same ADK agent graph, same ABAC policy model.
- **Many "on-prem vs GCP" choices collapse to a connection string + one-line index strategy** (`USING scann` vs `USING hnsw`) + per-backend ABAC compiler.
- **ABAC nuance:** policy model + orchestration reused; enforcement compiled per adapter. NOT identical SQL everywhere.

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Part 2 of 3

Deep Dive: Dimension by Dimension

Dimension: Time-to-Market

Platform assembly + hardening only (inference external both ways; CORE identical). Deltas are relative effort bars.

Capability	GCP	On-prem	Edge
API gateway / authz edge	light	heavy	GCP — Kong HA, plugin tuning, cert rotation are DIY
Guardrails / injection	light	heavy	GCP — NeMo rails authored/tested; host Llama Guard
Retrieval / index / rerank	light	heaviest	GCP — OpenSearch sizing + separate reranker + fusion
Doc parsing / layout	light	heavy	GCP — Layout/OCR hand-tuned; scale the parser yourself
PII / redaction	light	heavy	GCP — Presidio recognizers, custom entities, FP/FN tuning
Event-trigger ingestion	light	heavy	GCP — Event mesh, scale-to-zero, retry/DLQ self-built
Object store	light	medium	GCP — Distributed MinIO, erasure coding, lifecycle
Structured + vector DB	light	heavy	GCP — Self-run HA Postgres: replication, failover, PITR
Eval / observability	light	heaviest	GCP — Four OSS tools to deploy, wire, dashboard, upgrade
Cross-cutting security	light	heavy	GCP — More seams → IAM, network policy, pen-test
Total (assembly to prod)	~7 eng-weeks	~36 eng-weeks	GCP ~5× faster

Time-to-Market: Milestones

Milestone	GCP	On-prem	Edge
First working slice	~2 weeks	~6–8 weeks	GCP
Production-hardened (HA, security, eval, observability)	~6–8 weeks	~5–7 months	GCP
Ongoing platform ops (recurring)	low (config)	~0.5–1.0 FTE steady-state	GCP

- TTM total (~7 vs ~36 eng-weeks) is **one-time assembly only**. The ~0.5–1.0 FTE is the separate recurring ops line, not double-counted.
- The shared CORE refactor is the Year-0 build common to both; the TTM **delta** is platform assembly, not the CORE.
- On-prem also carries a recurring patch/upgrade tax across ~10 OSS systems plus four external inference endpoints.

Verdict: GCP wins decisively (~5×). Weeks vs quarters to a hardened platform. INDICATIVE (2026).

Dimension: Quality

Sub-dimension	GCP-managed	On-prem OSS	Edge
Retrieval (hybrid + rerank)	Agent Search on Gemini Enterprise Agent Platform: managed RRF + tuned reranker, improved by Google	OpenSearch BM25/kNN + bge-reranker, self-wired/tuned	GCP
Layout-aware parsing	Document AI: tables, forms, reading order, OCR managed	Tika/Unstructured: good text, weaker tables/scans without tuning	GCP
Model / generation	Gemini: ~1M context, native tools, strong grounding	Gemma (hosted): solid floor, trails on long-context + nested structured output	GCP
Guardrail coverage	Model Armor: managed injection + jailbreak + RAI + DLP-aware, auto-updated	Llama Guard 4 + NeMo + Presidio: strong, but you author/maintain rails	GCP
PII detection	Cloud DLP: 100+ infoTypes, checksum/context validation	Presidio: strong on regex PII; NER quality = your model + tuning	GCP
Eval tooling	Gemini Enterprise Agent Platform Evals: turnkey faithfulness/relevance/precision/recall	RAGAS + Langfuse + Phoenix: best-in-class OSS, but self-assembled	GCP (slight)

Quality: The Honest Read

- **Where GCP clearly wins:** reranker + layout-parser combo raises the retrieval floor with zero tuning; Gemini > Gemma on grounded generation; guardrail/DLP coverage stays current without rail-authoring labor.
- **Honest counter-point:** a well-tuned OpenSearch + bge-reranker-v2-m3 is genuinely strong; the retrieval gap narrows once tuned. OSS eval (RAGAS/Langfuse/Phoenix) is legitimately excellent.
- **The durable gap is in the MODEL (Gemini vs Gemma) and the zero-tuning FLOOR** — not in measurement. The model gap can't be closed by ops effort.
- **Cross-target caveat:** absolute eval scores stay backend-relative. Regression gates must use normalized metric names/ranges, not absolute numbers.

Verdict: GCP wins; gap is largest exactly where RAG answer quality is made.

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Dimension: Cost / TCO

Inference excluded (equal, external both sides). INDICATIVE annual run-rate (infra + people).

Cost shape	GCP-managed	On-prem-on-K8s	Edge
Pricing model	Usage-based opex; scales toward zero when idle	License + 3-yr hardware amortization + people	GCP
HA/DR, patching, certs, upgrades	Provider-absorbed	Engineered, staffed, and drilled in-house	GCP
Platform/SRE headcount	~1.0–1.5 FTE (integrate + observe)	~4–6 FTE (operate substrate 24×7 + on-call)	GCP
Annual run-rate (ex-inference)	~\$350k–\$700k	~\$1.1M–\$2.2M	GCP

- Headcount assumes **net-new hires**. If an existing under-utilized platform team absorbs it, the people delta shrinks materially (see "Where On-Prem Wins"). The two assumptions are mutually exclusive.
- On-prem also runs four external inference endpoints (LLM, embedder, reranker, safety); GCP consumes one model endpoint + the managed bundle.

Verdict: GCP wins. Headcount, not hardware, is the differentiator. INDICATIVE (2026).

Cost: Why the Headcount Gap Is Real

Each on-prem role is undifferentiated heavy lifting — none differentiates the product:

On-prem role	Why required
K8s / platform engineer(s)	Cluster lifecycle, upgrades, autoscaling, add-ons, capacity
Search / DB specialist	OpenSearch shard/JVM/heap tuning; pgvector index + failover
Storage / eventing engineer	MinIO erasure-coding + rebuilds; Knative/KEDA/Argo health
Security engineer	CVE patching cadence, secrets, policy, DLP/guardrail upkeep, audit
On-call rotation	24x7 needs ≥ 4 people to be humane and sustainable

On GCP the same coverage is the provider's job, baked into per-unit pricing and shared across the fleet. On-prem pays full loaded salary and can't share it. This is what "open source is free" comparisons leave out.

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3-Year TCO Summary

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Line	GCP-managed	On-prem-on-K8s
Year 0 one-time (build / migration)	\$0.15M	\$0.45M
Year 1 run-rate (infra + people)	\$0.53M	\$1.65M
Year 2 run-rate (+10% GCP usage; on-prem flat)	\$0.58M	\$1.68M
Year 3 run-rate	\$0.63M	\$1.72M
3-year TCO (ex-inference)	~\$1.9M	~\$5.5M
+ shared inference (~\$60k/yr × 3, equal both)	+\$0.18M	+\$0.18M
3-year fully-loaded TCO	~\$2.1M	~\$5.7M

- **GCP lands at ~35–40% of on-prem 3-year TCO.** Gap dominated by people (~\$0.75M cumulative on GCP vs ~\$3.6M on-prem).
- Year-0 delta is platform assembly; the shared CORE refactor is equal in both.

Dimension: Risk (Net Posture)

Net = exposure after typical mitigations.

Risk factor	GCP-managed	On-prem OSS	Edge
Operational (systems to run/scale/fail over)	Low (managed)	High (~10 on-call surfaces + 4 inference endpoints)	GCP
Security-patch lag (CVEs)	Low (provider-side)	High (you own CVE-to-patch SLA across the stack)	GCP
Talent / key-person	Low-Med (common skills)	High (scarce OpenSearch + PG-HA + K8s-eventing depth)	GCP
Supply-chain (deps / images)	Low (managed surfaces)	High (transitive deps across ~10 projects)	GCP
Cost overrun	Med (needs budgets/quotas/alerts)	Med (pre-provision peak; step-function)	Tie
Data residency / sovereignty	Low-Med (region pin, provider-held)	Low (full physical control)	On-prem

Verdict: GCP wins on net risk. Its top risks are policy- and architecture-governable; on-prem's top risks require continuous scarce labor. INDICATIVE (2026).

Dimension: Security & Compliance

Control	GCP-managed	On-prem OSS	Edge
Encryption / key management	CMEK via Cloud KMS, auto rotation, key-level audit	DIY KMS / Vault + KES	GCP
Network isolation	VPC Service Controls perimeter, IAM conditions, private endpoints	NetworkPolicy + mesh mTLS (SPIFFE/SPIRE) self-built	GCP
Inherited certifications	Provider compliance scope inherited (Assured Workloads)	Assemble + attest scanning, secrets, policy yourself	GCP
Edge protection	Cloud Armor managed WAF (OWASP CRS) + L7 DDoS at Google edge	ModSecurity/Coraza WAF; perimeter DDoS solved outside gateway	GCP
Audit logging	Cloud Audit Logs feed the CORE audit model	OTel/Prom/Grafana feed it with more wiring	GCP
ABAC invariant (server-side, every retrieval)	CORE policy model, compiled to Agent Search filter-DSL	CORE policy model, compiled to OpenSearch DLS / SQL	Tie (by design)
Dev-auth <code>X-User</code> bypass in prod	Apigee strips client-supplied identity headers	Gateway strips headers; trust only signed/mTLS identity	Tie (must-do both)

Verdict: GCP wins on built-in controls and inherited certification scope. INDICATIVE (2026).

Dimension: Data Residency / Sovereignty

Factor	GCP-managed	On-prem OSS	Edge
Physical location of data	Region-pinned, in provider infrastructure	Fully in customer DC/cluster; control plane can be air-gapped	On-prem
"Data never leaves org boundary" mandate	Region pin + VPC-SC + CMEK; ultimate control is the cloud's	Full control; satisfies the strictest mandates	On-prem
Inference egress (both targets)	External model call leaves the boundary	External model call leaves the boundary	Tie (by assumption)
Control-plane air-gap	Not possible (managed control plane)	Possible for the platform (model call still external)	On-prem

Key caveat: a true air-gap forbidding external model calls breaks **both** designs equally and needs a different inference story — a model hosted inside the sovereign boundary.

Verdict: On-prem wins. Strongest, often decisive on-prem case. A constraint that justifies on-prem, not a cost saving. INDICATIVE (2026).

Dimensions: Scalability & Ops

Dimension	GCP-managed	On-prem OSS	Edge
Elasticity / capacity	Autoscale + scale-to-zero; usage-based, smooth	KEDA/Knative help, but reserved platform nodes wasteful when idle	GCP
Durability (object store)	GCS 11-nines (multi/dual-region), strong consistency	MinIO erasure-coding matches on paper; realized = ops maturity	GCP
Vector-retrieval at scale	AlloyDB ScaNN (vendor-cited faster vs pgvector HNSW)	pgvector HNSW; solid, manual tuning required	GCP
Operational burden	None (no cluster); provider-absorbed patching	Continuous K8s + ~10 OSS + 4 inference endpoints to run/patch	GCP
Talent / skills	Common skills, ~1.0–1.5 FTE	Scarce depth, ~4–6 FTE	GCP

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The Decision

Weighted Scorecard (Transparent)

Scores 1–5 (5 = best). Weights sum to 97.

Dimension	Wt	GCP	On-prem	GCP wtd	On-prem wtd
Time-to-market	15	5	2	0.75	0.30
Quality (retrieval/guardrails/eval/model)	15	5	3	0.75	0.45
Cost / TCO (fully loaded)	15	5	2	0.75	0.30
Operational burden	12	5	2	0.60	0.24
Risk (net posture)	10	4	2	0.40	0.20
Security & compliance	8	5	3	0.40	0.24
Scalability / reliability	8	5	3	0.40	0.24
Observability / eval maturity	5	5	3	0.25	0.15
Talent / skills	5	4	2	0.20	0.10
Data residency / sovereignty	4	2	5	0.08	0.20
TOTAL	97			4.58	2.42

GCP 4.58 vs On-prem 2.42. GCP leads every high-weight dimension; on-prem leads only the lowest-weight one. INDICATIVE (2026).

Scorecard: Sovereignty as a Gate

The weighted score answers the *default* question. A hard residency mandate changes the question type.

Scenario	How sovereignty is treated	Result
No hard residency mandate	Sovereignty is one weighted factor (weight 4)	GCP wins (4.58 vs 2.42)
Hard residency mandate applies	Sovereignty is a pass/fail GATE , not a weighted factor	On-prem wins outright (GCP is off the table regardless of price)

- When data legally may not sit in a managed cloud, you don't weigh it — it's a gate, and only the on-prem profile clears it.
- Absent that gate, the weighted result is robust to reasonable weight changes.

Sensitivity check: $\text{gap} = 4.58 - 2.42 = \sim 2.16$. Doubling sovereignty weight (4→8) shifts $\sim +0.08$ to on-prem, leaving ~ 2.1 in GCP's favor. The conclusion holds.

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Risk Register (After Mitigation)

On-prem

Risk	Net	Mitigation
Operational (~10 systems + 4 inference endpoints)	High	Hard to mitigate without heavy FTE + maturity
Security-patch lag (OSS CVEs)	High	Patch cadence + scanning you operate
Talent / key-person	High	Cross-training; skills are scarce
Supply-chain (deps / image CVEs)	High	SBOM + image scanning you run

GCP-managed

Risk	Net	Mitigation
Cost overrun	Med	Billing budgets, quotas, alerts, Gemini Enterprise Agent Platform eval sampling
Data residency / sovereignty	Med	Region pin + VPC-SC + CMEK (else use the on-prem profile)
Service deprecation	Low	Provider lifecycle notices; adapter-isolated

Cross-cutting (both): ABAC in CORE so neither runtime can leak access control into the model; OTel is the portable telemetry seam; spoofable dev X-User must be hard-off in prod.

Honest Case: Where On-Prem Genuinely Wins

Driver	Why on-prem is the right call	Caveat
Hard data-sovereignty / residency mandate	Data + metadata legally must never leave the org boundary; region pinning is insufficient.	A constraint, not a saving. Justifies on-prem even at higher cost.
Air-gapped / classified control plane	No outbound connectivity to a public-cloud control plane permitted.	External inference is the KEY ASSUMPTION in both; a true air-gap forbidding model calls breaks both designs.
Existing idle datacenter capacity	Hardware capex is sunk; marginal cost is power + a cluster slice.	Must be genuinely idle AND adequate. "Idle but old" still fails.
Existing under-utilized platform team	Dominant cost (people) already paid with durable headroom; on-call already covers 24x7.	If you hire for this, the gap reopens immediately.
Pre-existing OpenSearch / Postgres / K8s estate	Marginal team + license cost is incremental, shared across workloads.	Only if this app is a small add to an already-funded platform.

Rule of thumb: on-prem breaks even mainly when hardware AND people are already sunk + shared, or when residency law removes GCP from contention. INDICATIVE (2026).

Rebuttal: Why GCP Still Nets Out Ahead

On-prem argument	Rebuttal
"Open source is free"	Inference is external both ways — on-prem's only saving is moot, while it carries ~4–6 FTE ops + four inference endpoints GCP bundles. 3-yr TCO ~\$2.1M vs ~\$5.7M.
"We control our data"	Region pin + VPC-SC + CMEK cover most cases; for a genuine sovereignty mandate, move THAT workload to the on-prem profile — same product, no rewrite.
"We can match the quality"	Only with sustained tuning; the model gap (Gemini vs Gemma) and managed reranker + layout-parser floor can't be closed by ops.
"We can build it"	Yes — in ~5× the eng-weeks, months later, then ~0.5–1.0 FTE forever to keep ~10 OSS systems + four inference endpoints healthy.

The clincher: the same hexagonal design that makes on-prem possible is what makes GCP safe. Portability is an architectural property, not a deployment choice. INDICATIVE (2026).

Portability Caveats (Stated Plainly)

The exit path is real but not frictionless. Honest leak points:

Area	The leak	Consequence
Object store API	GCS offers an S3-compatible API, but some advanced S3 features differ (V4 signing, generation-vs-opaque version IDs, multipart edge cases)	Small provider-specific branch in the storage adapter
ABAC enforcement	One policy model, compiled to different mechanisms (SQL / Agent Search filter-DSL / OpenSearch DLS)	Security-critical reimplementations per adapter, validated separately
Retrieval chunking	GCP may auto-chunk inside Agent Search on Gemini Enterprise Agent Platform; on-prem reuses CORE chunking	Chunk boundaries (and citation spans) not byte-identical across targets
Eval scores	Different judge models/prompts per backend	Absolute scores not comparable; gates use normalized names/ranges

Net: none of these flips the recommendation. They convert falsifiable claims into claims that survive scrutiny. INDICATIVE (2026).

Delivery Roadmap (Phased)

Each phase is shippable; the demo spine comes first. Shared CORE refactor is Year-0, common to both targets.

Phase	Ships	Profile	Milestone
0. Carve-out	CORE + ports + composition root; local adapters	local	Full demo runs on SQLite/FTS5/pypdf (119 tests)
1. GCP data + retrieval	AlloyDB, GCS, Document AI, Agent Search on Gemini Enterprise Agent Platform, Eventarc	gcp	ABAC-filtered ingest + reranked retrieval on GCP
2. GCP generation + edge	Gemini, Apigee / IAP, Cloud Observability + Gemini Enterprise Agent Platform Evals	gcp	Full GCP RAG behind Apigee, metrics on a dashboard
3. GCP safety	Model Armor, Cloud DLP	gcp	Injection blocked, PII redacted; GCP target complete
4. On-prem data + retrieval	pgvector, MinIO, Tika, OpenSearch + bge, Knative/KEDA	onprem	Same demo on K8s; ABAC pushdown validated
5. On-prem gen + safety + obs	Gemma (LiteLLM), Kong/OIDC, Llama Guard + NeMo + Presidio, OTel	onprem	Both targets pass the same conformance + eval suite
6. Compare + harden	Cross-target eval, cost model, runbooks, exit-path test	both	This deck, backed by measured numbers

Final Recommendation

- **Default to GCP-managed.** Wins every high-weight dimension: fastest TTM (weeks vs quarters), best quality (reranker + layout parsing + Gemini), lowest TCO (~35–40% of on-prem), most governable risk. Scorecard **4.58 vs 2.42**.
- **BUILD on-prem to a tested baseline as insurance** for genuine trump cards: a hard sovereignty/residency mandate, air-gap control, or an existing hardened platform + ops team you're obligated to reuse.
- **Govern GCP's real risks by policy:** budgets + quotas + alerts (cost), region pin + VPC-SC + CMEK (residency).

De-risking property	Mechanism
Product never gets rewritten	One provider-agnostic CORE (ABAC, chunking, retrieval, Validator, citations, audit, UI) serves both targets.
Switching cost is bounded	Only adapters + profile change; ABAC policy, retrieval, and audit invariants are shared.
Data is portable	AlloyDB is Postgres-wire-compatible; GCS offers an S3-compatible API.
Telemetry is portable	OTel SDK in CORE; swap the exporter, not the instrumentation.
Sovereign workloads relocate later	Start on GCP for speed + quality; move a regulated workload to on-prem when a mandate demands — no product rewrite.

Bottom line: with one portable CORE, choose GCP-managed now for the fastest, highest-quality, lowest-TCO path — knowing the architecture preserves a clean exit to on-prem for any future sovereignty obligation. All figures INDICATIVE (2026).